

Is Progressive Disclosure
All You Need for Long-Context Agents?

EMNLP
2026

Yifeng He¹ **Yinzhe Zhao^{2*}** Jicheng Wang¹ Hao Chen³

¹ University of California, Davis ² Zhejiang University ³ The University of Hong Kong



Disclosure gains depend on scale, task and harness.

Flat often helps at library scale; hierarchy helps in some settings and hurts in others.

01 Three routes to the same evidence

RAW

No prepared skill metadata

↓

Search and read the raw book

↓

Read matched passages

FLAT

book-nav /

1: always in context
SKILL.md (frontmatter)
name: book-nav
description: Navigate Frankenstein
by Mary Shelley to answer questions.

2: loaded on activation

Instructions to use this Skill

Table of Contents
ch-01	Walton, Arctic
ch-02	Victor's youth
...	...
3: loaded on demand
references/
chapter-01.md
chapter-02.md
...
chapter-31.md

always in context on activation on demand

HIERARCHICAL

book-nav /

1: always in context
SKILL.md (meta / router)
name: book-navigation
description: Navigate Frankenstein
(31 chapters, each a skill)

2: loaded on activation

Instructions to use these chapter Skills

chapter-01 /

1: always in context
SKILL.md (frontmatter)
name: book-chapter-01
description: Robert Walton writes from
the Arctic; rescues Victor from the
ice.

2: loaded on activation
SKILL.md body:
Letter 1 - To Mrs. Saville, England.
You will rejoice to hear that ...

... chapter-31 /

always in context on activation on demand

One skill per book indexes all chapters.

Only the book description stays in context.

Open the skill to load the chapter index.

Read selected chapter files on demand.

One book router plus one skill per chapter.

All chapter descriptions stay in context.

Each skill body contains its chapter text.

More routing depth and more initial context.

LOONGDOC ∞Bench book QA in a sandboxed file system; tool trajectories and scores logged.

3 harnesses: Codex, Pi, Claude-Code 3 model families En.MC / En.QA / Zh.QA 1, 5, 10, 20 books

02 Single book: all harnesses, models and tasks

En.MC = English multiple choice En.QA = English open QA Zh.QA = Chinese open QA

Codex	Pi			Pi	Claude-Code		
	En.MC	En.QA	Zh.QA		En.MC	En.QA	Zh.QA
Raw	0.8943	0.7412	0.8652	Raw	0.8851	0.7161	0.6856
Flat	0.8977	0.7516	0.8390	Flat	0.9126	0.7259	0.7007
Hier.	0.8874	0.7377	0.8525	Hier.	0.6398	0.5120	0.6510

Pi	Claude-Code		
	En.MC	En.QA	Zh.QA
Raw	0.7865	0.6813	0.7563
Flat	0.8023	0.6913	0.7479
Hier.	0.6460	0.5933	0.3890

Claude-Code	Pi		
	En.MC	En.QA	Zh.QA
Raw	0.7448	0.6892	0.8142
Flat	0.8667	0.7173	0.8330
Hier.	0.8687	0.7255	0.8295

Gains depend on the setup

Flat adds little on Codex, but improves English MC more on Claude-Code / Haiku.

Hierarchy can hurt

On Pi / GPT, English MC falls from 0.9126 (Flat) to 0.6398 (Hierarchical).

03 Library scale: all tasks at 5, 10 and 20 books

All heatmaps share a 0–1 color scale. Darker = higher score. Bold = highest mean within a setting, not statistical significance.

Codex				Claude-Code				Pi			
gpt-5.4-mini				claude-haiku-4.5				gpt-5.4-mini			
	5 books	10 books	20 books		5 books	10 books	20 books		5 books	10 books	20 books
En.MC				En.MC				En.MC			
Raw	0.818	0.767	0.720	Raw	0.745	0.653	0.459	Raw	0.410	0.391	0.333
Flat	0.752	0.790	0.760	Flat	0.781	0.636	0.482	Flat	0.470	0.434	0.386
Hier.	0.777	0.791	0.746	Hier.	0.777	0.655	0.521	Hier.	0.463	0.386	0.299
En.QA				En.QA				En.QA			
Raw	0.657	0.577	0.257	Raw	0.582	0.413	0.301	Raw	0.392	0.276	0.128
Flat	0.708	0.582	0.462	Flat	0.642	0.480	0.354	Flat	0.432	0.221	0.184
Hier.	0.667	0.606	0.267	Hier.	0.589	0.453	0.302	Hier.	0.364	0.310	0.168
Zh.QA				Zh.QA				Zh.QA			
Raw	0.698	0.214	0.043	Raw	0.730	0.550	0.499	Raw	0.510	0.414	0.287
Flat	0.738	0.178	0.095	Flat	0.696	0.576	0.401	Flat	0.378	0.283	0.137
Hier.	0.771	0.137	0.097	Hier.	0.687	0.533	0.389	Hier.	0.505	0.307	0.330

Flat helps English QA

Across 5, 10 and 20 books, Flat exceeds Raw in all six Codex and Claude-Code English-QA settings.

Hierarchy has exceptions

On Pi Chinese QA, Hier. exceeds Flat at every scale. At 20 books: 0.330 vs. 0.137.

No universal winner

On Claude-Code Chinese QA, Raw leads at 5 and 20 books. Choose by task and scale, not by structure alone.

Start with Flat, then validate on your task and scale

Flat beats Raw in all six English-QA settings for Codex and Claude-Code at library scale. Hierarchy is not uniformly worse: on Pi Chinese QA it exceeds Flat at all three library sizes.

Scope: narrative books in ∞Bench. English MC may be affected by memorized novels; code/manual generalization is untested.

Source: Tables 1–3, Sections 3–5 and Appendix C.1. * Work done while interning at UC Davis. Supported by UC Noyce Initiative.

